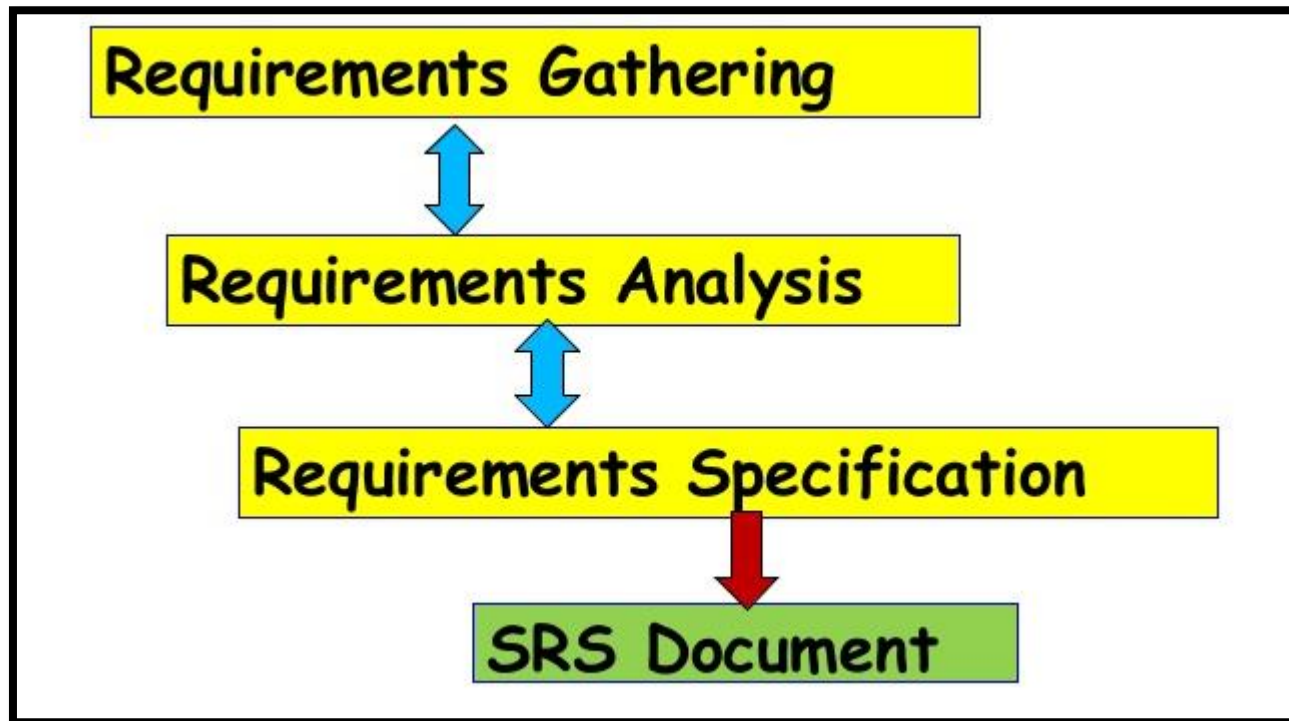


Requirements Analysis and Specification



Requirements Analysis and Specification

- ▶ Many projects fail:
 - ▶ because they start implementing the system without determining whether they are building what the customer really wants.





How the customer explained it



How the project leader understood it



How the engineer designed it



How the programmer wrote it



How the sales executive described it



How the project was documented



What operations installed



How the customer was billed



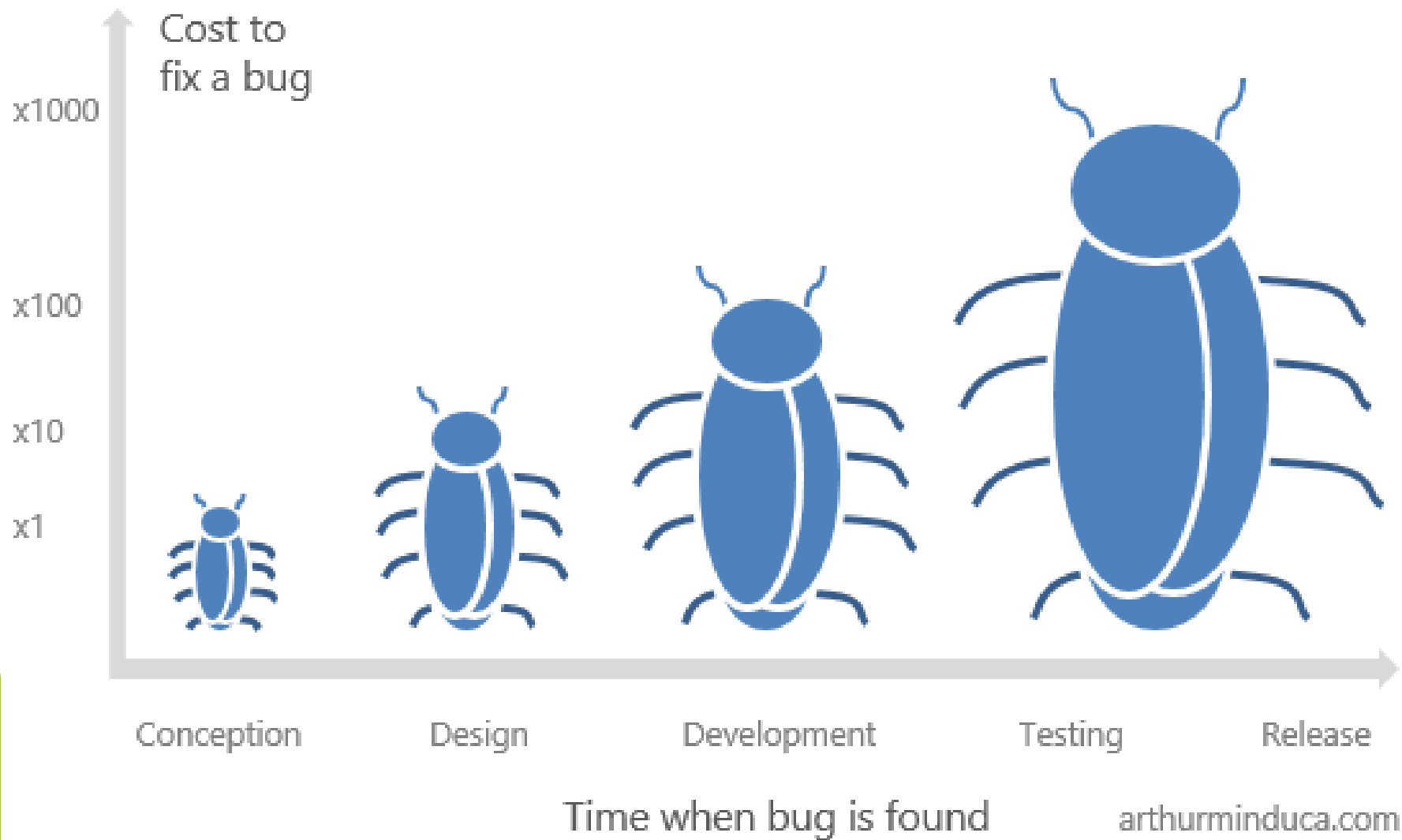
How the helpdesk supported it



What the customer really needed

Requirements Analysis and Specification

- ▶ Goals of requirements analysis and specification phase:
 - ▶ fully understand the user requirements
 - ▶ remove inconsistencies, anomalies, etc. from requirements
 - ▶ document requirements properly in an SRS document



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Requirements Analysis and Specification

- ▶ Consists of two distinct activities:
 - ▶ Requirements Gathering and Analysis
 - ▶ Specification

- ▶ Analyst gathers requirements through:
 - ▶ observation of existing systems,
 - ▶ studying existing procedures,
 - ▶ discussion with the customer and end-users,
 - ▶ analysis of what needs to be done, etc.





Requirements Gathering (CONT.)

- ▶ In the absence of a working system,
 - ▶ lot of imagination and creativity are required.
- ▶ Interacting with the customer to gather relevant data:
 - ▶ requires a lot of experience.

Requirements Gathering (CONT.)

- ▶ Some desirable attributes of a good system analyst:
 - ▶ Good interaction skills,
 - ▶ imagination and creativity,
 - ▶ experience.



Analysis of the Gathered Requirements

- ▶ After gathering all the requirements:
 - ▶ analyze it:
 - ▶ Clearly understand the user requirements,
 - ▶ Detect inconsistencies, ambiguities, and incompleteness.
- ▶ Incompleteness and inconsistencies:
 - ▶ resolved through further discussions with the end-users and the customers.

Inconsistent requirement

- ▶ Some part of the requirement:
 - ▶ contradicts with some other part.

▶ Example:

- ▶ One customer says turn off heater and open water shower when temperature $> 100^{\circ}\text{C}$
- ▶ Another customer says turn off heater and turn ON cooler when temperature $> 100^{\circ}\text{C}$



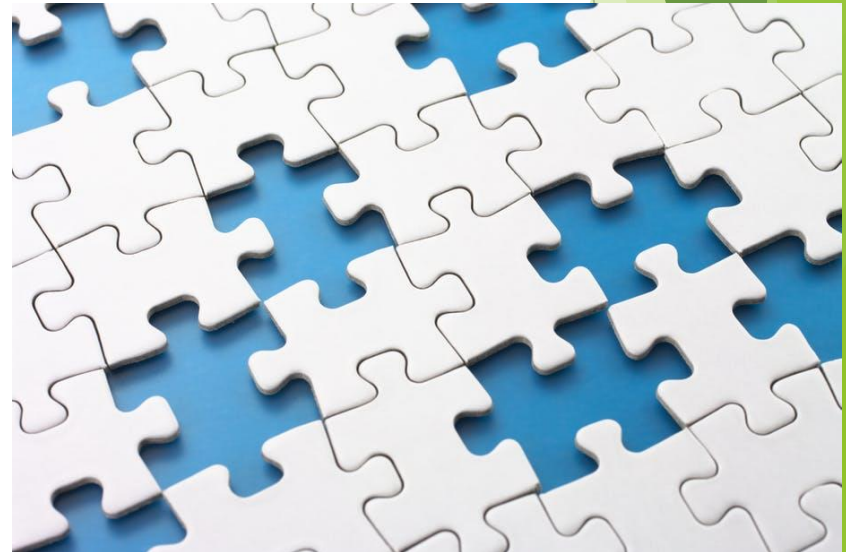
Incomplete requirement

- ▶ Some requirements have been omitted:
 - ▶ due to oversight.

- ▶ Example:

- ▶ The analyst has not recorded:
when temperature falls below 90 C
 - ▶ heater should be turned ON
 - ▶ water shower turned OFF.

INCOMPLETE



Analysis of the Gathered Requirements

(CONT.)

- ▶ Requirements analysis involves:
 - ▶ obtaining a clear, in-depth understanding of the product to be developed,
 - ▶ remove all ambiguities and inconsistencies.



Analysis of the Gathered Requirements_(CONT.)

- ▶ Several things about the project should be clearly understood by the analyst:
 - ▶ What is the problem?
 - ▶ Why is it important to solve the problem?
 - ▶ What are the possible solutions to the problem?
 - ▶ What complexities might arise while solving the problem?

Analysis of the Gathered Requirements_(CONT.)

- ▶ After collecting all data regarding the system to be developed,
 - ▶ remove all inconsistencies and anomalies from the requirements,
 - ▶ systematically organize requirements into a Software Requirements Specification (SRS) document.



Requirements Analysis and Specification

- ▶ The person who undertakes requirements analysis and specification:
 - ▶ known as **systems analyst**:
 - ▶ collects data pertaining to the product
 - ▶ analyzes collected data:
 - ▶ to understand what exactly needs to be done.
 - ▶ writes the **Software Requirements Specification (SRS)** document.

**SOFTWARE
REQUIREMENT
SPECIFICATION**

Requirements Analysis and Specification

- ▶ Final output of this phase:
 - ▶ Software Requirements Specification (SRS) Document.
- ▶ The SRS document is reviewed by the customer.
 - ▶ reviewed SRS document forms the basis of all future development activities.

SRS Template

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1. Introduction

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5. Other Nonfunctional Requirements

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Appendix A: Glossary

Appendix B: Analysis Models

Appendix C: Issues List

Software Requirements Specification

- ▶ Main aim of requirements specification:
 - ▶ systematically organize the requirements arrived during requirements analysis
 - ▶ document requirements properly.





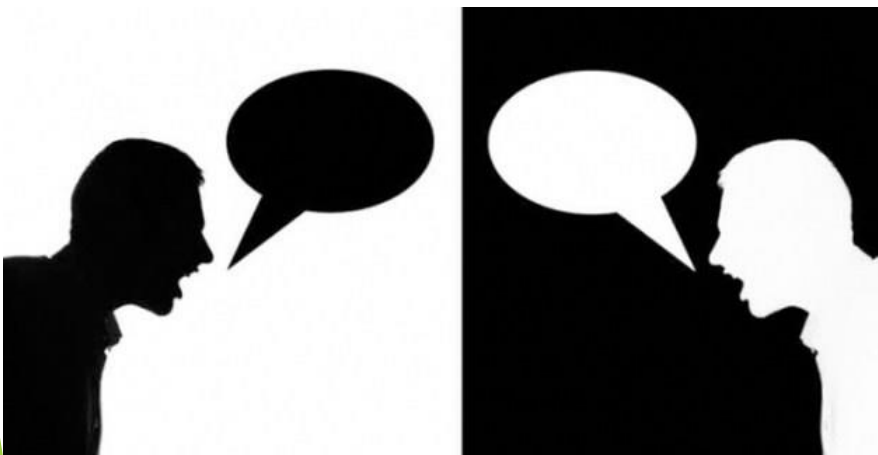
Software Requirements Specification

- ▶ The SRS document is useful in various contexts:
 - ▶ statement of user needs
 - ▶ contract document
 - ▶ reference document
 - ▶ definition for implementation



Software Requirements Specification: A Contract Document

- ▶ Requirements document is a reference document.
- ▶ SRS document is a contract between the development team and the customer.
- ▶ Once the SRS document is approved by the customer,
 - ▶ any subsequent controversies are settled by referring the SRS document.



Software Requirements Specification: A Contract Document

- ▶ Once customer agrees to the SRS document:
 - ▶ development team starts to develop the product according to the requirements recorded in the SRS document.
- ▶ The final product will be acceptable to the customer:
 - ▶ as long as it satisfies all the requirements recorded in the SRS document.

SRS Document (CONT.)

- ▶ The SRS document is known as black-box specification:
 - ▶ the system is considered as a black box whose internal details are not known.
 - ▶ only its visible external (i.e. input/output) behaviour is documented.



1. Introduction
 - 1.1 Purpose of the SRS document
 - 1.2 Scope of the document
 - 1.3 Definitions, acronyms, abbreviations used in the document
 - 1.4 References
 - 1.5 Overview of the document
2. General description
 - 2.1 Product description giving the overall product features
 - 2.2 Functionality of the product
 - 2.3 User characteristics (types of users, such as administrators, operators)
 - 2.4 General constraints in the development of the product
 - 2.5 Assumptions and dependencies (assumptions made in the preparation of the SRS and the external systems on which the software is dependent)
3. Functional requirements
 - 3.1 Introduction
 - 3.2 Inputs
 - 3.3 Outputs
 - 3.4 Processing details
4. External interface requirements
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 - 5.1 Product requirements
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 - 5.1.2 Portability
 - 5.1.3 Usability
 - 5.1.4 Performance requirements
 - 5.2 Process requirements
 - 5.2.1 Delivery schedule
 - 5.2.2 Hardware/Software requirements
 - 5.2.3 Standards compliance (whether the software has to conform to any national or international standards)
 - 5.3 External requirements
 - 5.3.1 Safety requirements
 - 5.3.2 Privacy/Security requirements

SRS Document (CONT.)



- ▶ SRS document concentrates on:
 - ▶ what needs to be done
 - ▶ carefully avoids the solution (“how to do”) aspects.
- ▶ The SRS document serves as a contract
 - ▶ between development team and the customer.
 - ▶ Should be carefully written

SRS Document (CONT.)

- ▶ The requirements at this stage:
 - ▶ written using end-user terminology.
- ▶ later a formal requirement specification may be developed from it.

Properties of a good SRS document

- ▶ It should be concise
 - ▶ and at the same time should not be ambiguous.
- ▶ It should specify what the system must do
 - ▶ and not say how to do it.
- ▶ Easy to change.,
 - ▶ i.e. it should be well-structured.
- ▶ It should be consistent.
- ▶ It should be complete.

Properties of a good SRS document (cont...)

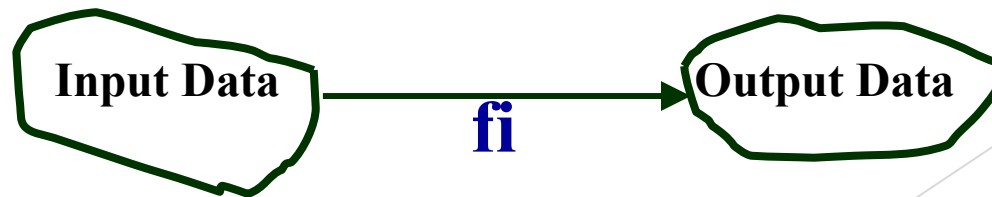
- ▶ It should be traceable
 - ▶ you should be able to trace which part of the specification corresponds to which part of the design and code, etc and vice versa.
- ▶ It should be verifiable
 - ▶ e.g. “system should be user friendly” is not verifiable

SRS Document (CONT.)

- ▶ SRS document, normally contains three important parts:
 - ▶ functional requirements,
 - ▶ Non functional requirements,
 - ▶ constraints on the system.

SRS Document (CONT.)

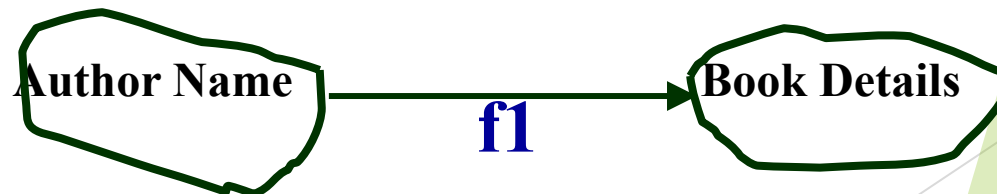
- ▶ It is desirable to consider every system:
 - ▶ performing a set of functions $\{f_i\}$.
 - ▶ Each function f_i considered as:
 - ▶ transforming a set of input data to corresponding output data.



Example: Functional Requirement

► F1: Search Book

- Input:
 - an author's name:
- Output:
 - details of the author's books and the locations of these books in the library.



Functional Requirements

- ▶ Functional requirements describe:
 - ▶ A set of high-level requirements
 - ▶ Each high-level requirement:
 - ▶ takes in some data from the user
 - ▶ outputs some data to the user
 - ▶ Each high-level requirement:
 - ▶ might consist of a set of identifiable functions

Functional Requirements

- ▶ For each high-level requirement:
 - ▶ every function is described in terms of
 - ▶ input data set
 - ▶ output data set
 - ▶ processing required to obtain the output data set from the input data set

Nonfunctional Requirements

- ▶ Characteristics of the system which can not be expressed as functions:
 - ▶ maintainability,
 - ▶ portability,
 - ▶ usability, etc.

Nonfunctional Requirements

- ▶ Nonfunctional requirements include:
 - ▶ reliability issues,
 - ▶ performance issues,
 - ▶ human-computer interface issues,
 - ▶ Interface with other external systems,
 - ▶ security, maintainability, etc.

Constraints

- ▶ Constraints describe things that the system should or should not do.
 - ▶ For example,
 - ▶ standards compliance
 - ▶ how fast the system can produce results
 - ▶ so that it does not overload another system to which it supplies data, etc.

Examples of constraints

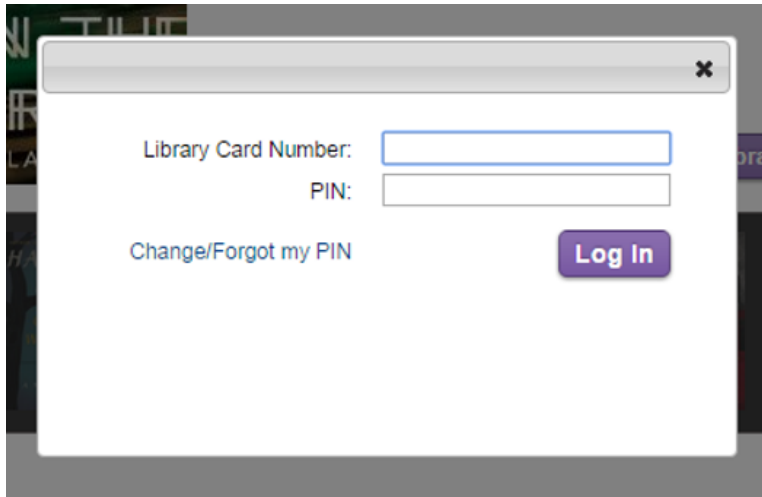
- ▶ Hardware to be used,
- ▶ Operating system
 - ▶ or DBMS to be used
- ▶ Capabilities of I/O devices
- ▶ Standards compliance
- ▶ Data representations
 - ▶ by the interfaced system

Organization of the SRS Document

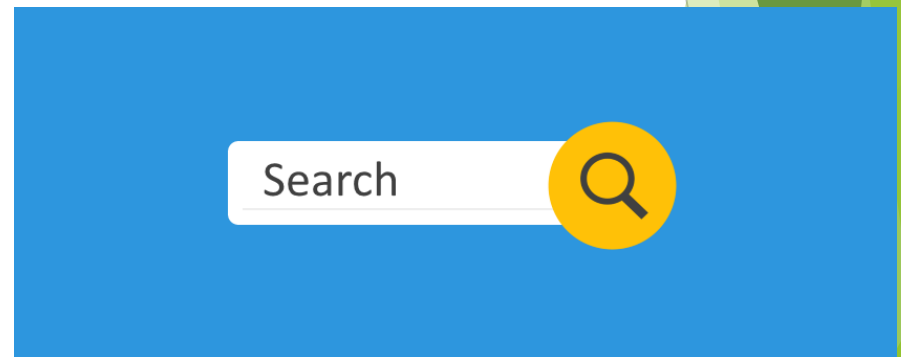
- ▶ Introduction.
- ▶ Functional Requirements
- ▶ Nonfunctional Requirements
 - ▶ External interface requirements
 - ▶ Performance requirements
- ▶ Constraints

Example Functional Requirements

- ▶ List all functional requirements
 - ▶ with proper numbering.
- ▶ Req. 1:
 - ▶ Once the user selects the “search” option,
 - ▶ he is asked to enter the key words.
 - ▶ The system should output details of all books
 - ▶ whose title or author name matches any of the key words entered.
 - ▶ Details include: Title, Author Name, Publisher name, Year of Publication, ISBN Number, Catalog Number, Location in the Library.



A screenshot of a library login form. It features two input fields: "Library Card Number:" and "PIN:". Below the PIN field is a link that says "Change/Forgot my PIN". To the right of these fields is a purple button labeled "Log In". The form is displayed in a window with a standard title bar and a close button (X) in the top right corner.



Example Functional Requirements

- ▶ **Req. 2:**
 - ▶ When the “renew” option is selected,
 - ▶ the user is asked to enter his membership number and password.
 - ▶ After password validation,
 - ▶ the list of the books borrowed by him are displayed.
 - ▶ The user can renew any of the books:
 - ▶ by clicking in the corresponding renew box.

Select Some or All Items to Renew

User ID:

or

Member No.

PIN:

List Charged Items

Reset Query Values

Req. 1:

▶ R.1.1:

- ▶ **Input:** “search” option,
- ▶ **Output:** user prompted to enter the key words.

▶ R1.2:

- ▶ **Input:** key words
- ▶ **Output:** Details of all books whose title or author name matches any of the key words.
 - ▶ Details include: Title, Author Name, Publisher name, Year of Publication, ISBN Number, Catalog Number, Location in the Library.
- ▶ **Processing:** Search the book list for the keywords

Req. 2:

▶ R2.1:

- ▶ **Input:** “renew” option selected,
- ▶ **Output:** user prompted to enter his membership number and password.

▶ R2.2:

- ▶ **Input:** membership number and password
- ▶ **Output:**
 - ▶ list of the books borrowed by user are displayed. User prompted to enter books to be renewed or
 - ▶ user informed about bad password
- ▶ **Processing:** Password validation, search books issued to the user from borrower list and display.

Req. 2:

► R2.3:

- **Input:** user choice for renewal of the books issued to him through mouse clicks in the corresponding renew box.
- **Output:** Confirmation of the books renewed
- **Processing:** Renew the books selected by the in the borrower list.

Examples of Bad SRS Documents

▶ Unstructured Specifications:

- ▶ Narrative essay --- one of the worst types of specification document:

- ▶ Difficult to change,
- ▶ difficult to be precise,
- ▶ difficult to be unambiguous,
- ▶ scope for contradictions, etc.

▶ Forward References:

- ▶ References to aspects of problem
 - ▶ defined only later on in the text.

Examples of Bad SRS Documents

▶ Overspecification:

- ▶ Addressing “how to” aspects
- ▶ For example, “Library member names should be stored in a sorted descending order”
- ▶ Overspecification restricts the solution space for the designer.

▶ Contradictions:

- ▶ Contradictions might arise
 - ▶ if the same thing described at several places in different ways.